

Programming Assignment

1. What is a Program?

A program is a set of instructions written in a programming language that tells a computer what to do.

Hello World in C:

```
#include  
  
int main() {  
    printf("Hello, World!");  
    return 0;  
}
```

Hello World in Python:

```
print("Hello, World!")
```

2. What is Programming?

Programming is the process of writing and maintaining code to create software.

Steps:

- Problem understanding
- Algorithm design
- Coding
- Testing
- Debugging
- Maintenance

3. High-level vs Low-level Languages:

High-level: Easy, portable (Python, Java)

Low-level: Hard, machine-dependent (Assembly)

4. Internet Working:

Client -> Internet -> Server -> Response

5. TCP/IP Layers:

Application, Transport, Internet, Network

6. Client-Server:

Client requests, server responds.

7. Internet Types:

Broadband, Fiber, Satellite

8. HTTP vs HTTPS:

HTTP is not secure, HTTPS is secure.

9. Security:

SQL Injection, XSS, Weak Passwords

10. Software Types:

System, Application, Utility

11. Architecture:

Presentation -> Logic -> Database

12. Environments:

Development, Testing, Production

13. Source Code vs Machine Code:

Readable vs Binary

14. GitHub:

Version control system.

15. SDLC:

Requirement, Design, Development, Testing, Deployment, Maintenance